

## Anaerobic and aerobic energy production of young rainbow trout (*Salmo gairdneri*) during and after bursts of activity

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Accepted January 10, 1985

**Summary.** 1. Anaerobic and aerobic energy production during and after induced bursts of maximum swimming activity were measured in four size classes of rainbow trout fry at 4, 12 and 20 °C.

2. The rate of phosphocreatine (PCr) utilization during burst activity appeared independent of size and temperature. On average the concentration of PCr dropped from  $5.8 \pm 1.0$  to  $1.5 \pm 0.7 \mu\text{mol} \cdot \text{g}^{-1}$  within 30 s of stimulation, and decreased only slightly more after an additional 30 sec of stimulation. The concentration of ATP also decreased to 70% of control levels, whereas the concentrations of ADP and AMP remained constant, in agreement with other investigations on vertebrate white muscles. Lactate concentrations did not change during the shorter period of stimulation but from then on increased steadily through the second half of the one minute stimulus and through the recovery period observed (10 min).

3. Five minutes after the end of stimulation phosphocreatine had recovered 67% of the control level whereas the concentration of ATP as well as the energy charge remained low. After ten minutes not only phosphocreatine but also ATP and energy charge had recovered at least 80% of the ground lost during the period of exertion.

4. The following parameters of aerobic energy production were distinguished: *Routine rate* ( $\dot{V}_{\text{O}_2\text{r}}$ ): maintained for several hours before or after induced burst activity; *maximum rate* ( $\dot{V}_{\text{O}_2\text{max}}$ ): attained for a few minutes during and immediately after an induced burst of activity; *net recovery*  $V_{\text{O}_2}$ : amount of oxygen consumed during recovery time minus the amount due to routine rate during this interval.

*Symbols and abbreviations:* PCr phosphocreatine;  $\dot{V}_{\text{O}_2\text{r}}$  routine rate of oxygen consumption;  $\dot{V}_{\text{O}_2\text{max}}$  maximum rate of oxygen consumption

5. The routine rate of  $\dot{V}_{\text{O}_2}$  is affected by temperature but less so by the age of the fry. On the other hand,  $\dot{V}_{\text{O}_2\text{max}}$  and net recovery  $V_{\text{O}_2}$  increased strongly with age at the two lower temperatures. Aerobic scope for activity increases with size and is maximum at 12 °C.

6. The age-dependent changes of  $\dot{V}_{\text{O}_2}$  are discussed in relation to the differentiation of the swimming musculature and to hydrodynamic changes, since between a body length of 1 and 10 cm, the fish move through a range of Reynolds numbers from approximately  $10^2$  to  $10^4$ .

7. It appears, furthermore, that with increasing size, more energy is liberated during the recovery period than would be required for repaying the true oxygen debt. This may be due to an elevated activity during poststimulus excitement.

### Introduction

The energy metabolism of fish has been thoroughly investigated in the past and there exist many excellent reviews (Fry 1971; Beamish 1978; Brett and Groves 1979). However, most of these investigations were concerned with adults or at least with fairly large fish. Both the hydrodynamic and the metabolic aspects of swimming in very young fish have been neglected in the literature (see Blake 1983).

There are two reasons why we may expect differences between young and adult fish in the use of energy during locomotion. One is that after hatching the swimming musculature develops only gradually and, besides, in a species specific manner. Both in *Coregonus* sp. (Forstner et al. 1983) and in *Salmo gairdneri* (Nag and Nursall 1972),

the red and pink muscles do not begin to differentiate until several weeks after hatching. On the other hand, a thin layer of red muscle is present right after hatching in *Coregonus* sp. but not in young rainbow trout. Thus, in fry of the former species, the activities of the enzymes of the citric acid cycle are high compared with those of glycolysis but low in those of the latter species (Hinterleitner and Wieser, to be published).

The second reason why one may expect differences between the energetics of young and adult fish is that the two size classes swim in quite different ranges of Reynolds numbers. A fish measuring 20 cm in length, swimming at 1 body length  $\cdot$  s<sup>-1</sup>, has a Reynolds number of approximately  $4 \times 10^4$ , one measuring 1 cm and swimming at the same relative speed, one of  $10^2$  (Childress 1981). Thus, viscous forces may be expected to play a much larger role in swimming fry than in swimming adults (Webb 1975). In order to pursue some of these questions, we have started a program investigating various aspects of swimming and energy metabolism in very young fish.

In this paper we analyze the energy metabolism of rainbow trout of different ages during induced bursts of maximum swimming activity. On the basis of what has been said above, we may expect a gradual development of metabolic responses of the fry as they increase in size and as their muscles differentiate.

## Material and methods

**Animals.** All experiments were carried out on rainbow trout, the eggs of which were obtained from a fish breeding station near Innsbruck (Institut für Fischforschung, Thaur) or from a hatchery in Italy (Preore). Fry and juveniles of different ages, from yolk sac stage to six months, were used for the measurements of either anaerobic or aerobic energy production. Before the experiments, the fish were maintained in running water of approximately 8 °C and after absorption of the yolk were fed dry food (Trouvit). The following size classes will be discerned: Size class I: yolk sac stage to beginning of free swimming phase (80–120 mg); size class II: 120–200 mg; size class III: 200–1,000 mg; size class IV: 3–10 g.

**Experimental procedure for determination of anaerobic energy production.** Individual fry (from ca. 80 to 1,000 mg) were subjected for either 30 or 60 sec to either electrical or mechanical (chasing with a glass rod) stimulation in small jars submerged in water of constant temperature. No differences, resulting from the different stimuli, were noted. Electrical stimulation was with rectangular pulses of 5 ms width, the voltage and frequency of which varied between 1 and 25 V and 1 and 10 Hz respectively. The animals appeared much more active during the first than during the second 30 sec which is borne out by the rate of phosphagen breakdown (see below).

Changes in energy status before, during and after stimulation were followed by measuring the concentrations of ATP, ADP, AMP, phosphocreatine (PCr) and lactate in the whole

fish. Experiments were carried out at 4, 12 and 20 °C, to which temperatures the animals were transferred 24 h prior to the experiments.

**Extraction and determination of metabolites.** Individual fish were quickly grasped with forceps, blotted and dropped into liquid propane at liquid nitrogen temperature. In the later stages of this work, fish were freeze-clamped at liquid N<sub>2</sub> temperature. No differences were observed to result from the two methods. After freezing, the fish were blotted again, weighed and then ground with 1 ml of frozen trichloroacetic acid (TCA, 7% in distilled water w/v) under liquid nitrogen. The powder was transferred to ice-cold 7% TCA, so that the final dilution of the tissues was 4- to 5-fold, thoroughly mixed, centrifuged at 12,000 g for 30 min, the supernatant neutralized with 5 M KHCO<sub>3</sub> and centrifuged again. The metabolites were determined enzymatically using the following standard spectrophotometric procedures: ATP, ADP and AMP according to Jaworek, Gruber and Bergmeyer; phosphocreatine according to Lamprecht et al.; L-lactate according to Gutmann and Wahlefeld, all to be found in Bergmeyer (1974). 100 µl of neutralized extract were added to 0.9 ml of assay medium.

**Respirometry:** Oxygen consumption was measured in a flow-through system with an oxygen-sensitive electrode. The system was submerged in a thermostated water bath. By turning a lever it could be closed so that water circulated through the cylindrical animal chamber and past the oxygen electrode (see Forstner 1983). A chamber with a volume of 149 ml was used for 5–10 fry weighing up to 200 mg, or for 1–2 fry weighing 500 to 1,000 mg each. A chamber with a volume of 761 ml was used for larger fish (up to 10 g). Flow rates through the chambers varied between approximately 200 and 400 ml  $\cdot$  min<sup>-1</sup>.

The fish were not allowed to extract more than 20% of the oxygen content of the water volume, after which a further turn of the lever permitted the system to be flushed with air-saturated water from the surrounding bath. The experimental system was kept in the open position overnight, the fish remaining inside without food. The animal chamber was equipped with two metal grids connected to a stimulator, this permitting electrical stimulation of the fish without interruption of the measurement of oxygen consumption. The oxygen electrode was not affected by the electrical fields created by the stimulator. The stimuli were comparable to those used for the determination of anaerobic energy production and lasted for exactly 60 s.

Bacterial oxygen consumption was measured at the beginning and at the end of the experiment. A correction was applied to fish respiration assuming a linear increase of bacterial oxygen consumption from beginning to end.

## Results

### Variability of energy status during spontaneous activity

The rainbow trout used in this study were hatched and raised in the laboratory in slowly running water of approximately 8 °C. Under these conditions the yolk sac disappears 25 to 35 days after hatching. During the last week or so of this stage the fry begin to swim freely, assuming a very typical locomotor pattern in which short spurts of swimming alternate with longer phases of main-

**Table 1.** Effects of short-term electrical stimulation and recovery on energy status of young rainbow trout. Summary of experiments conducted at 4, 12 and 20 °C with fry weighing from ca. 80 mg to 1.0 g. Whole fish were analyzed; *n*=number of independent measurements, each measurement representing one fish. Since within the investigated range neither size nor temperature had a significant effect on the concentration of the metabolites all experimental results were pooled

|                 | AMP                            | ADP  | ATP  | Total<br>adeny-<br>lates | Energy<br>charge               | PCr | Lactate |
|-----------------|--------------------------------|------|------|--------------------------|--------------------------------|-----|---------|
|                 | [μmol (g f.w.) <sup>-1</sup> ] |      |      |                          | [μmol (g f.w.) <sup>-1</sup> ] |     |         |
| Controls        |                                |      |      |                          |                                |     |         |
| $\bar{x}$       | 0.12                           | 0.23 | 1.91 | 2.26                     | 0.90                           | 5.8 | 1.45    |
| <i>s</i>        | 0.05                           | 0.13 | 0.62 |                          | 0.03                           | 1.0 | 1.00    |
| <i>n</i>        | 23                             | 23   | 23   | 23                       | 23                             | 23  | 41      |
| Stimulus 30 s   |                                |      |      |                          |                                |     |         |
| $\bar{x}$       | 0.12                           | 0.26 | 1.40 | 1.78                     | 0.86                           | 1.5 | 1.40    |
| <i>s</i>        | 0.05                           | 0.05 | 0.65 |                          | 0.04                           | 0.7 | 0.80    |
| <i>n</i>        | 18                             | 18   | 18   | 18                       | 18                             | 18  | 21      |
| Stimulus 60 s   |                                |      |      |                          |                                |     |         |
| $\bar{x}$       | 0.12                           | 0.32 | 1.34 | 1.78                     | 0.83                           | 1.2 | 2.20    |
| <i>s</i>        | 0.03                           | 0.10 | 0.38 |                          | 0.05                           | 0.4 | 1.00    |
| <i>n</i>        | 16                             | 16   | 16   | 16                       | 16                             | 16  | 10      |
| Recovery 5 min  |                                |      |      |                          |                                |     |         |
| $\bar{x}$       | 0.11                           | 0.26 | 1.34 | 1.71                     | 0.81                           | 3.9 | 3.40    |
| <i>s</i>        | 0.04                           | 0.08 | 0.43 |                          | 0.02                           | 1.0 | 1.30    |
| <i>n</i>        | 18                             | 18   | 18   | 18                       | 18                             | 18  | 8       |
| Recovery 10 min |                                |      |      |                          |                                |     |         |
| $\bar{x}$       | 0.11                           | 0.27 | 1.66 | 2.04                     | 0.87                           | 4.7 | 3.80    |
| <i>s</i>        | 0.04                           | 0.08 | 0.42 |                          | 0.04                           | 1.2 | 1.10    |
| <i>n</i>        | 12                             | 12   | 12   | 12                       | 12                             | 12  | 10      |

taining position against the gentle currents. During these phases the fry are almost stationary above ground, which they achieve with seemingly little effort. The energy status – that is, the levels of key metabolites involved in energy metabolism – of a fish may depend on its activity prior to being captured. We conjectured that burst swimming entails an increased rate of glycolysis and thus expected to find large variations in the lactate concentrations of fish removed at random from the holding tanks. As shown in Table 1, this turned out to be the case since the coefficient of variation for lactate (67%) is large, much larger, for example, than that for phosphocreatine (17%).

#### *Anaerobic energy production during induced burst activity*

When forced to maximum exertion the rate of anaerobic energy production in the fry was indepen-

**Table 2.** PCr concentrations after 30 s of stimulation and after 5 min of recovery in fry of rainbow trout at different temperatures. Means  $\pm$  SD (*n*). Concentrations in the controls and after 30 s of stimulation are not significantly different at the three experimental temperatures

|                | Phosphocreatine (μmol·g <sup>-1</sup> ) |                   |                   |
|----------------|-----------------------------------------|-------------------|-------------------|
|                | 4 °C                                    | 12 °C             | 20 °C             |
| Control        | 6.0 $\pm$ 0.7 (8)                       | 5.7 $\pm$ 0.8 (9) | 6.0 $\pm$ 1.0 (6) |
| Stimulus 30 s  | 1.4 $\pm$ 0.4 (7)                       | 1.9 $\pm$ 0.9 (7) | 1.3 $\pm$ 0.6 (6) |
| Recovery 5 min | 3.3 $\pm$ 1.1 (7)                       | 4.1 $\pm$ 0.6 (8) | 4.9 $\pm$ 0.9 (3) |

dent of age and size within the range investigated. Even sac fry showed the same responses as freely swimming fish of 1 g. The data as summarized in Table 1 permit the following conclusions:

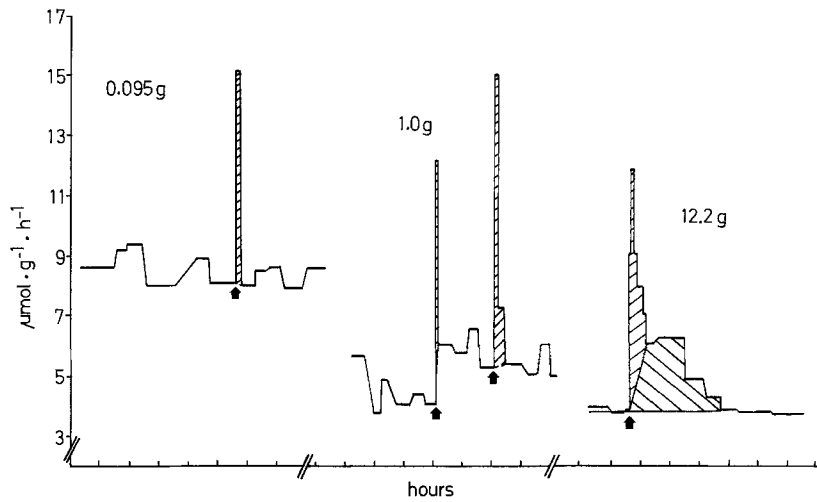
a) Within 30 s of stimulation the PCr stores of the body have become all but exhausted, decreasing only slightly more after an additional 30 s of stimulation.

b) The concentration of ATP also decreases to 70% of control levels within 60 s. However, the concentrations of ADP and AMP remain constant, indicating that the AMP deaminase occurring in the muscles of adult fish (Markarewicz 1969; Driedzic and Hochachka 1976; Raffin and Leray 1980; van Waarde and Kesbeke 1981) is already very active in the fry.

c) Lactate remained constant during the first 30 s period of stimulation, but between 30 and 60 s its concentration increased from 1.40 $\pm$ 0.8 to 2.2 $\pm$ 1.0 μmole·g<sup>-1</sup> (*P*<0.05, *n*=10).

#### *Recovery*

The major sign of recovery is the return of the concentrations of PCr to control levels. Five minutes after the end of stimulation PCr had increased from 1.2 $\pm$ 0.4 to 3.9 $\pm$ 1.0 μmol·g<sup>-1</sup>, and after 10 min the concentration reached 4.7 $\pm$ 1.2 μmol·g<sup>-1</sup>, only slightly lower than the pre-stimulus level. There was no relation to size on the time course of PCr regeneration. Anaerobic glycolysis may be partly responsible for the reconstitution of PCr during the first 5 min of recovery. The increase from 2.2 to 3.4 μmol lactate·g<sup>-1</sup> corresponds to the generation of 1.8 μmoles of ATP from glycogen as against the 2.7 μmoles required for the synthesis of PCr during this time. Energy charge and the concentration of ATP remained at their lowest points during the first five minutes of recovery, but after 10 min control levels were nearly regained.



**Fig. 1.** Illustration of the effect of a one minute electrical stimulus on the rate of oxygen consumption ( $\mu\text{mol}\cdot\text{g}^{-1}\cdot\text{h}^{-1}$ ) in three size classes of rainbow trout. Recovery periods marked by hatched columns. In the largest fish recovery was particularly protracted and appeared to consist of two phases, a fast and a slow one, which is indicated by different shading. Experimental temperature was 12 °C

**Table 3.** Routine (r) and maximum (max) rates of oxygen consumption ( $\mu\text{mol}\cdot\text{g}^{-1}\cdot\text{h}^{-1}$ ) in four size classes of rainbow trout at three experimental temperatures. Means  $\pm$  SD ( $n$  of independent measurements)

| Size class | 4 °C                           |                                  | 12 °C                          |                                  | 20 °C                          |                                  |
|------------|--------------------------------|----------------------------------|--------------------------------|----------------------------------|--------------------------------|----------------------------------|
|            | $\dot{V}_{\text{O}_2\text{r}}$ | $\dot{V}_{\text{O}_2\text{max}}$ | $\dot{V}_{\text{O}_2\text{r}}$ | $\dot{V}_{\text{O}_2\text{max}}$ | $\dot{V}_{\text{O}_2\text{r}}$ | $\dot{V}_{\text{O}_2\text{max}}$ |
| I          | $2.5 \pm 0.4$<br>$n=4$         | $6.6 \pm 0.9$<br>$n=4$           | $4.8 \pm 1.1$<br>$n=5$         | $8.9 \pm 1.6$<br>$n=5$           |                                |                                  |
| II         | $3.0 \pm 0.4$<br>$n=5$         | $7.8 \pm 0.7$<br>$n=5$           | $8.6 \pm 2.0$<br>$n=7$         | $15.3 \pm 2.9$<br>$n=7$          | $14.0 \pm 1.1$<br>$n=3$        | $26.0 \pm 3.0$<br>$n=3$          |
| III        | $3.8 \pm 1.2$<br>$n=9$         | $11.5 \pm 3.9$<br>$n=9$          | $7.1 \pm 1.7$<br>$n=7$         | $18.7 \pm 5.2$<br>$n=7$          | $18.0 \pm 4.3$<br>$n=7$        | $29.3 \pm 7.3$<br>$n=7$          |
| IV         | $3.0 \pm 0.5$<br>$n=8$         | $15.4 \pm 2.3$<br>$n=8$          | $6.3 \pm 1.2$<br>$n=7$         | $30.1 \pm 15.6$<br>$n=7$         | $11.3 \pm 1.3$<br>$n=4$        | $22.2 \pm 2.4$<br>$n=4$          |

**Table 4.** Three respiratory relationships based on rates of oxygen consumption of four size classes of rainbow trout as summarized in Table 3. *Net recovery* oxygen demand ( $\mu\text{mol}\cdot\text{g}^{-1}$ ) = amount of oxygen consumed after stimulation until a new steady rate is achieved; *aerobic scope* ( $\mu\text{mol}\cdot\text{g}^{-1}\cdot\text{h}^{-1}$ ) =  $\dot{V}_{\text{O}_2\text{max}} - \dot{V}_{\text{O}_2\text{r}}$ ; *factorial scope* =  $\dot{V}_{\text{O}_2\text{max}}/\dot{V}_{\text{O}_2\text{r}}$ . Means  $\pm$  SD

| Size class | 4 °C                                   |                |                 | 12 °C                                  |                |                 | 20 °C                                  |               |                 |
|------------|----------------------------------------|----------------|-----------------|----------------------------------------|----------------|-----------------|----------------------------------------|---------------|-----------------|
|            | Net recovery<br>$\dot{V}_{\text{O}_2}$ | Aerobic scope  | Factorial scope | Net recovery<br>$\dot{V}_{\text{O}_2}$ | Aerobic scope  | Factorial scope | Net recovery<br>$\dot{V}_{\text{O}_2}$ | Aerobic scope | Factorial scope |
| I          | 0.35<br>(0.50)                         | 4.15<br>(0.70) | 2.7<br>(0.4)    | 0.41<br>(0.18)                         | 4.1<br>(0.7)   | 1.9<br>(0.2)    | —                                      | —             | —               |
| II         | 0.84<br>(0.40)                         | 4.9<br>(0.3)   | 2.7<br>(0.2)    | 0.74<br>(0.21)                         | 6.7<br>(1.2)   | 1.8<br>(0.2)    | 1.05<br>(0.01)                         | 12.0<br>(1.9) | 1.8<br>(0.1)    |
| III        | 1.34<br>(0.53)                         | 7.9<br>(2.7)   | 3.0<br>(0.4)    | 1.09<br>(0.33)                         | 11.6<br>(3.8)  | 2.6<br>(0.3)    | 1.5<br>(0.4)                           | 11.3<br>(4.2) | 1.6<br>(0.2)    |
| IV         | 2.00<br>(0.46)                         | 12.4<br>(2.1)  | 5.2<br>(0.8)    | 3.18<br>(1.60)                         | 23.8<br>(14.7) | 4.6<br>(1.8)    | 2.7<br>(1.0)                           | 11.1<br>(1.2) | 2.0<br>(0.1)    |

### Temperature relationship of anaerobic energy production

No effect of temperature – between 4 and 20 °C – could be detected on the concentrations of the compounds studied or on the rates of phosphagen breakdown, the most important source of anaerobic energy during the first 30 s of burst swimming (Table 2). The resynthesis of PCr appeared to proceed more slowly at 4 than at 20 °C but at the latter temperature the number of samples was small. Other changes connected with stimulation, like the slight decreases in ATP concentration and energy charge, also proved to be temperature-independent. For this reason data from different temperatures were pooled in Table 1.

### Aerobic energy production

**Definition of levels of oxygen consumption.** After they had been in the respiration chamber for about 24 h, the fish usually maintained a fairly constant rate of oxygen consumption. However, the variability of this rate between individuals or between groups of fish was high, amounting to 100% and more at each temperature. When the fish were stimulated to burst activity, oxygen consumption rose sharply to a *maximum rate* ( $\dot{V}_{O_2\max}$ ) for a few minutes and then returned either to the prestimulus level or to a higher one which sometimes was maintained for many hours. With the system used it was not possible to resolve the exact onset of  $\dot{V}_{O_2\max}$  but it began either within or immediately after the stimulus period and always lasted longer than the duration of the stimulus. The pattern of oxygen consumption is illustrated by the recordings for three specimens (or groups of specimens) of different weight measured at 12 °C (Fig. 1). In two of the fish (or groups of fish) the  $\dot{V}_{O_2}$  returned to the original level after stimulation, in the larger of the two after a very extended period of recovery. However, in the specimen weighing 1 g, a rate of oxygen consumption 25–50% above the pre-stimulus rate was maintained for four hours, although it was interrupted by a second period of forced burst swimming.

These 'steady' rates before and after stimulation will be lumped under the term *routine rate* of oxygen consumption ( $\dot{V}_{O_2r}$  in  $\mu\text{mol O}_2 \cdot \text{g}^{-1} \cdot \text{h}^{-1}$ ). *Recovery  $\dot{V}_{O_2}$*  is defined as the amount of oxygen consumed after stimulation until the rate has returned either to the prestimulus level (Fig. 1, 95 mg fish) or to a second, elevated rate maintained for several hours (Fig. 1, 1.0 g fish, first peak). *Net recovery  $\dot{V}_{O_2}$*  ( $\mu\text{mol O}_2 \cdot \text{g}^{-1}$ ) is the

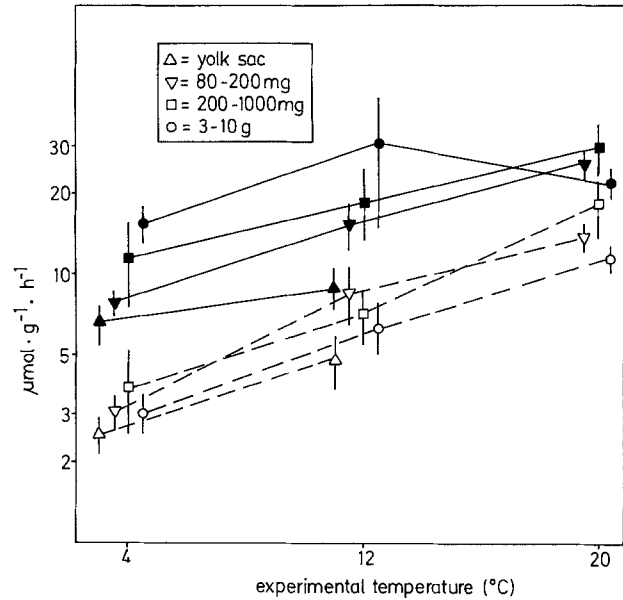


Fig. 2. Temperature dependence of log oxygen consumption in four size classes (inset) of rainbow trout. Open symbols: routine rate; closed symbols: maximum rate of oxygen consumption ( $\mu\text{mol} \cdot \text{g}^{-1} \cdot \text{h}^{-1}$ ). Mean values  $\pm$  SD

amount of oxygen consumed during *recovery time* minus the 'steady' rate. If the 'steady' rate before and after stimulation was different the arithmetic mean between the two was subtracted from total recovery  $\dot{V}_{O_2}$ .

In addition to the three size classes of fish examined for anaerobic energy production, a group of larger fish, defined as class IV (see Material and methods) was also included for the determination of aerobic energy production.

**Temperature relationships and age dependence of aerobic energy production.** Tables 3 and 4 show that the routine  $\dot{V}_{O_2}$  is strongly affected by temperature but less so by the age of the fish. There is the expected increase from sac fry to class II, but between the latter and class IV there is little difference in weight specific  $\dot{V}_{O_2r}$ . On the other hand,  $\dot{V}_{O_2\max}$  increased strongly with age and size at the two lower temperatures. At 12 °C  $\dot{V}_{O_2\max}$  of class IV fish was more than three times that of class I. At 20 °C however, a ceiling seems to have been reached and the effect of size on  $\dot{V}_{O_2\max}$  is insignificant (fry at the yolk sac stage could not be measured at this temperature). Temperature effects on  $\dot{V}_{O_2r}$  and  $\dot{V}_{O_2\max}$  are compared in Fig. 2.

Net recovery  $\dot{V}_{O_2}$  as well as recovery time (not shown here) increased with  $\dot{V}_{O_2\max}$ . Thus the greater the capacity for oxygen consumption the greater the burden that maximum swimming activity appears to impose on the body of the fish.

**Table 5.**  $Q_{10}$  – values for routine and maximum rates of oxygen consumption in four size classes (I, 80–120 mg; II, 120–200 mg; III, 200–1,000 mg; IV, 3–10 g) of juvenile *S. gairdneri*, based on measurements of oxygen consumption at 4, 12 and 20 °C

|                    |         | Temperature range - |         |
|--------------------|---------|---------------------|---------|
|                    |         | 4–12 °              | 12–20 ° |
| $\dot{V}_{O_2r}$   | Class I | 1.9                 | —       |
|                    | II      | 2.9                 | 1.8     |
|                    | III     | 1.9                 | 2.5     |
|                    | IV      | 2.3                 | 1.8     |
| $\dot{V}_{O_2max}$ | Class I | 1.3                 | —       |
|                    | II      | 2.0                 | 1.9     |
|                    | III     | 1.6                 | 1.6     |
|                    | IV      | 1.9                 | 0.7     |

Temperature effects on oxygen consumption were conventional but there was a tendency for  $Q_{10}$  – values to be lower for  $\dot{V}_{O_2max}$  than for  $\dot{V}_{O_2r}$  and to be higher between 4 and 12 °C than between 12 and 20 °C (Table 5).

What Fig. 2 and the data of Table 4 show is, in effect, the development of the scope for activity in young rainbow trout after hatching. This scope can be expressed as the *difference* between  $\dot{V}_{O_2max}$  and routine rate (Fry 1947) or as a *quotient*, the ‘factorial scope for activity’. As is evident from Table 4, there is a definite trend for both expressions to increase with size, and for aerobic scope to reach maximum values at the intermediate temperature.

## Discussion

The concentrations of energy-rich phosphates in young rainbow trout are somewhat lower than those of adult salmonids. In the white muscles of *Salvelinus fontinalis*, Walesby and Johnston (1980) measured between 5 and 6  $\mu\text{mol} \cdot \text{g}^{-1}$  of ATP and between 16 and 25  $\mu\text{mol} \cdot \text{g}^{-1}$  of PCr. Similar concentrations have been reported in eleven estuarine teleosts by Vetter and Hodson (1982). Since the white muscles of trout make up about 50% of the total body weight (Bone 1978) the highest concentrations summarized in Table 1 ought to correspond to average concentrations in the white muscles of 4–5 and 12–14  $\mu\text{mol} \cdot \text{g}^{-1}$  of ATP and PCr, respectively.

The major finding of the present investigation is that in young rainbow trout the *aerobic* scope for activity increases with age and size, whereas the rate of *anaerobic* energy production is unaffected by age or size of the fry. As in the locomotor muscles of other vertebrates, the first response to

stimulation is the breakdown of PCr. At the same time, the concentration of ATP declines to approximately 70% of the pre-stimulus value. In this way an average of 9.6  $\mu\text{mol ATP} \cdot \text{min}^{-1} \cdot \text{g}^{-1}$  is generated during the first 30 s of stimulation. This rate of anaerobic energy production is independent of fish size, at least from yolk sac stage to approximately 1 g, and of temperature between 4 and 20 °C, thus confirming previous reports on the temperature independence of burst swimming (Brett 1964). As in the white muscles of adult fish (van Waarde and Kesbeke 1981, 1983) and in the fast-twitch muscles of mammals (Meyer and Terjung 1979) a very active AMP deaminase might cooperate with myokinase in the swimming muscles of the fry to prevent the build-up of higher concentrations of ADP and AMP.

During recovery the regeneration of the PCr pool appears to follow a temperature-independent time course, 67% of control levels being attained after 5 min, 81% after 10 min. These figures correspond almost exactly to those of the gastrocnemius muscle of the rat recovering after tetanic stimulation at 37 °C (Meyer and Terjung 1979). Further agreement between fish and mammals exists with regard to the delay in the regeneration of the ATP pool. As in the rat there is no change in the concentration of ATP in the fish during the first five minutes after extreme exercise, although by that time phosphocreatine has regained two-thirds of its resting concentration. There may be significance in the observation by Onnen and Zebe (1983) that in the tail muscle of an active *invertebrate*, the shrimp *Crangon crangon*, ATP concentrations also decrease during heavy work but are back to normal within five minutes, whereas it takes much longer for the phosphagen, phosphoarginine, to return to resting levels. This difference between the two groups of animals may have something to do with the distribution of the enzyme AMP-deaminase which is very active in vertebrate, but absent from muscles of many invertebrates (Chandrasena and Hird 1978).

In contrast to anaerobic energy production, the aerobic scope for maximum activity develops gradually with the size of the fish and is strongly dependent on temperature (at least between 4 and 12 °C). Whereas the routine rate of oxygen consumption is little affected by size between 80 mg and 10 g fish, the scope for maximum activity increases three-fold at 4, and more than five-fold at 12 °C. At 20 °C, however, the routine rate of metabolism is so high that size has little additional effect on maximum aerobic activity.

This development of aerobic power is most

probably linked to the differentiation of the red and pink muscles, a process which in rainbow trout (Nag and Nursall 1972), but also in coregonids (Forstner et al. 1983), begins several weeks after hatching and takes several more weeks to be completed. After hatching the fry of rainbow trout with their large yolk sacs usually lie on the bottom of the tank and make only short excursions into the open water. It is not until 25 to 35 days later that they start swimming in a sustained manner which calls for a more versatile locomotor apparatus. On the other hand, the differentiation of the swimming musculature can also be seen in relation to the shift of Reynolds numbers as the fish grow longer. The fry of size class I measure about 1 cm in length, those of class IV 10 cm. Thus, swimming one body length  $\cdot \text{sec}^{-1}$  the fish move through a range of Reynolds numbers from approximately  $10^2$  to  $10^4$  which means that the ratio of inertial to viscous forces acting on the body changes by the same amount. This implies that the smaller fish need much less energy for acceleration than the larger fish (Webb 1975).

A word of caution ought to be added concerning the term 'scope for activity'. This term, as introduced by Fry (1947), has usually been taken to express the difference between oxygen consumption during *maximum sustained* swimming and 'resting' metabolism. In our experiments neither of the two conditions is met. The lower level of the scope for activity as defined in this paper is the lowest stable rate maintained for several hours before and after stimulation. This routine rate varies approximately two-fold even at one and the same temperature and without the fish showing any obvious differences in activity. Such a high degree of variability is usual in this group of animals (Fry 1971; Dolinin 1973). The highest level of oxygen consumption in our experiments is caused by stimulating the fish to a one-minute burst of maximum activity. However, the main driving force of this burst is likely to be anaerobic and oxygen consumption may be largely classified as 'recovery'. Thus one should be careful in extrapolating from the present findings to the energetics of long-term swimming activity in young fish. What the present investigation shows is the development of energy consuming responses of rainbow trout during and after short bursts of activity.

#### Replenishment of energy stores

The true oxygen 'debt' during recovery can be calculated with the data of Tables 1, 3 and 4. The cost of recovery consists, firstly, of phosphocrea-

**Table 6.** Energy 'demands' and 'supplies' (in  $\mu\text{mol ATP} \cdot \text{g}^{-1}$ ) in rainbow trout fry recovering from 1 min bursts of maximum activity. Three size classes of fry swimming at three temperatures are compared (see Material and methods). The definitions of 'demand' and 'supply' are the following: Demand: ATP required for replenishment of energy stores spent during burst activity; Supply: ATP supplied by anaerobic and aerobic processes during period of recovery

| Size class | 4 °C    |         | 12 °C   |         | 20 °C   |         |
|------------|---------|---------|---------|---------|---------|---------|
|            | De-mand | Sup-ply | De-mand | Sup-ply | De-mand | Sup-ply |
| I          | 6.5     | 5.9     | 6.5     | 6.3     | —       | —       |
| II         | 6.5     | 9.1     | 6.5     | 8.4     | 6.5     | 10.4    |
| III        | 6.5     | 12.3    | 6.5     | 10.7    | 6.5     | 12.1    |

tine and ATP regeneration which requires approximately  $5.2 \mu\text{mol ATP} \cdot \text{g}^{-1}$  at all temperatures and body sizes, and secondly of recharging the depleted oxygen stores of the body. If we assume a blood volume of 5% of body weight and an oxygen capacity of 10 volume % (Randall 1970)  $5 \mu\text{l}$  of oxygen would be present in a 1.0 g fish, equivalent to  $0.22 \mu\text{mol O}_2$  and  $1.36 \mu\text{mol ATP}$ . As usual in fish the oxidation of lactate is a slow process (Black 1957), thus during the first 15 min almost certainly no lactate is converted to glycogen. A total of  $6.56 \mu\text{mol ATP} \cdot \text{g}^{-1}$  is required by all size classes at all temperatures for the replenishment of energy stores. This is synonymous with the true oxygen debt and will be called energy *demand* in Table 6. The energy required for recovery is supplied by glycolysis and respiration. About  $1.6 \mu\text{mol lactate} \cdot \text{g}^{-1}$  are produced during the first 10 min of recovery at all temperatures and by all sizes, equivalent to  $2.4 \mu\text{mol ATP} \cdot \text{g}^{-1}$ . At the same time varying amounts of oxygen are consumed in excess of the pre-stimulus level by the different size classes of fry at the three experimental temperatures. If it is assumed that glycogen is the source of this energy  $1 \mu\text{mol O}_2$  is equivalent to  $6.5 \mu\text{mol ATP}$ . Thus total *supply* of energy during the recovery period consists of the ATP equivalents of lactate production ( $1 \mu\text{mol lactate} = 1.5 \mu\text{mol ATP}$ ) and of the ATP equivalents of what in Table 4 has been listed as 'net recovery' (in  $\mu\text{mol O}_2 \cdot \text{g}^{-1}$ ). In this way 'demands' and 'supplies' of energy during the recovery period can be calculated for the three size classes (excluding size class IV in which anaerobic energy production was not determined) at the three temperatures. The result of such a calculation (Table 6) shows that in the smallest fry the amount of ATP required for the replenishment of energy stores spent during activity, is well balanced against the amount of ATP supplied by anaerobic

and aerobic processes. However, with increasing size an increasing amount of energy is supplied in excess of what would be required for repaying the true oxygen debt. This excess energy must then have been used for covering additional costs of metabolism probably due to post-stimulus excitement. As in mammalian physiology 'recovery  $\dot{V}_{O_2}$ ' need not to be the same as 'O<sub>2</sub> debt' (Hagberg et al. 1980).

**Acknowledgement.** This work was supported by the 'Fonds zur Förderung der wissenschaftlichen Forschung in Österreich', project no. 4674.

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